Covariant Phase Space

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Abstract

Notes on covariant phase space with boundaries, mostly following the notes of [?].

Contents

1 Geometric Mechanics

A typical treatment of Hamiltonian mechanics takes a phase space plus a privileged choice of coordinates q^a, p^a and a scalar function H encoding dynamics as

$$\dot{q}^a = \frac{\partial H}{\partial p^a}, \quad \dot{p}^a = -\frac{\partial H}{\partial q^a}. \tag{1}$$

Equivalently, we take any (possibly explicitly time-dependent) observable $A = A(q(t), p(t), t)$ to time-evolve as

$$\begin{aligned} \frac{dA}{dt} &= \frac{\partial A}{\partial t} + \frac{\partial A}{\partial q^a}\frac{dq^a}{dt} + \frac{\partial A}{\partial p^a}\frac{dp^a}{dt} \\ &= \frac{\partial A}{\partial t} + \frac{\partial A}{\partial q^a}\frac{\partial H}{\partial p^a} - \frac{\partial A}{\partial p^a}\frac{\partial H}{\partial q^a} \\ &\equiv \frac{\partial A}{\partial t} + \{A, H\} \end{aligned} \tag{2}$$

where we have introduced the Poisson bracket

$$\{A, B\} \equiv \frac{\partial A}{\partial q^a}\frac{\partial B}{\partial p^a} - \frac{\partial A}{\partial p^a}\frac{\partial B}{\partial q^a} \tag{3}$$

and where (here and elsewhere) we will use the Einstein summation convention on index pairs.

This formalism is not incorrect, but in the present note we will prefer to work in the formalism of *geometric mechanics*, which makes symmetries and other abstract or geometric properties manifest by lifting the usual description to a coordinate-free one. To begin with, we will take our phase space to be a real 2-manifold $\mathcal{P}$. Rather than specifying a scalar function and a bracket outright, we encode the dynamics by specifying a *symplectic form* Ω, *i.e.*, a closed non-degenerate 2-form over $\mathcal{P}$. Abstractly, at any point $p \in \mathcal{P}$ we have

$$\Omega_p : T_p\mathcal{P} \times T_p\mathcal{P} \to \mathbb{R}, \quad (X_p, Y_p) \mapsto \Omega_p(X_p, Y_p) \tag{4}$$

If we put a vector in, say, the second slot and leave the first open, the resulting object takes in vectors and returns scalars.

$$\Omega_p(\cdot, Y_p) : T_p\mathcal{P} \to \mathbb{R}, \quad X_p \mapsto \Omega_p(X_p, Y_p). \tag{5}$$

So we have cooked up a 1-form. We can thus think of Ω equivalently as a map which takes in vectors and spits out 1-forms, and by mild abuse of notation we will also call this map Ω. Briefly, we have

$$\Omega(Y)(X) \equiv \Omega(X, Y). \tag{6}$$

Notice that the typographical ordering of X, Y is flipped across this equality. This is purely convention, but will matter for local minus signs.

The assumption of non-degeneracy exactly ensures that there exists an inverse map Ω^{-1} that takes 1-forms to vectors, or equivalently takes in two 1-forms and returns a scalar, or equivalently is a bivector.

$$\Omega^{-1}(\omega, \sigma) \in \mathbb{R}, \quad \omega\left(\Omega^{-1}(\sigma)\right) \equiv \Omega^{-1}(\omega, \sigma). \tag{7}$$

This is an inverse in the literal sense that

$$\Omega^{-1}(\Omega(X)) = X, \quad \Omega(\Omega^{-1}(\sigma) = \sigma. \tag{8}$$

This inverse map will be our coordinate-free implementation of a Poisson. Concretely, given any scalar function $H : \mathcal{P} \to \mathbb{R}$, we can define the vector field X_H on $\mathcal{P}$ by the rule

$$X_H(f) \equiv \Omega^{-1}(\delta f, \delta H). \tag{9}$$

Here we write δ for the exterior derivative on phase space, to disambiguate from the exterior derivative on spacetime. We define more generally

$$\{f, g\} \equiv \Omega^{-1}(\delta f, \delta g). \tag{10}$$

Then Ω (by way of Ω^{-1}) defined the dynamics of our theory as

$$\frac{df}{dt} = X_H(f) = \{f, H\}. \tag{11}$$

We can write this just as well in terms of Ω. First, take the 1-form $\delta f, \delta g$ and use Ω^{-1} to cast them to vectors.

$$X_f \equiv \Omega^{-1}(\delta f), \quad X_g \equiv \Omega^{-1}(\delta g) \tag{12}$$

or equivalently

$$\delta f = \Omega(X_f), \quad \delta g = \Omega(X_g). \tag{13}$$

Now we carefully manipulate all of the above definitions to find

$$\begin{aligned}
\{\delta f, \delta g\} &= \Omega^{-1}(\delta f, \delta g) \\
&= (\delta f)\left(\Omega^{-1}(\delta g)\right) \\
&= (\Omega(X_f))\left(\Omega^{-1}(\Omega(X_g))\right) \\
&= (\Omega(X_f))(X_g) \\
&= \Omega(X_g, X_f).
\end{aligned} \tag{14}$$

We can recover the usual story in coordinates by taking

$$\Omega = \delta p^a \wedge \delta q^a. \tag{15}$$

Darboux's theorem guarantees that we can always locally pick coordinates which diagonalize Ω in this sense.

1.1 Zero Modes and Gauge Symmetry

It may happen that we write down some candidate phase space $\tilde{\mathcal{P}}$ and corresponding symplectic form $\tilde{\Omega}$ and find that the latter is degenerate, or equivalently has zero-modes $\tilde{X}$. That is, there exists a nonzero vector field $\tilde{X}$ such that

$$\tilde{\Omega}(\tilde{X}, Y) = 0 \text{ for all } Y. \tag{16}$$

To be sure, our story above requires that the symplectic form be invertible independently at every point in phase space, so degeneracy at even one point would be enough to throw a wrench into our business. Here we only assume that $\tilde{X}$ isn't uniformly vanishing, but it doesn't need to be everywhere nonzero.

In this case we will be more conservative: we will call $\tilde{\mathcal{P}}$ the *pre-phase space* and $\tilde{\Omega}$ the *pre-symplectic form*. The canonical fix is then to quotient by these zero-modes, producing a quotiented phase space and a corresponding symplectic form which is non-degenerate by construct. More concretely, we identify any two points $\tilde{p}, \tilde{q} \in \tilde{\mathcal{P}}$ if they are connected by an integral curve of any 0-mode $\tilde{X}$.

(nice picture of integral curves and orbits)

We will think of these integral curves as gauge orbits, the 0-modes as generators of gauge transformations, and flowing along the curves as applying a gauge transformation $\tilde{g} \in \tilde{\mathcal{G}}$. Then our physical phase space is exactly the collection of gauge orbits

$$\mathcal{P} = \tilde{\mathcal{P}}/\tilde{G} \tag{17}$$

and by construction $\tilde{\Omega}$ induces a well-defined a non-degenerate symplectic form Ω on $\mathcal{P}$. Of course this all assumes the relevant quotient structure is analytically nice enough that $\mathcal{P}$ is itself a smooth manifold; this isn't guaranteed, but moving forward we will assume it happens.

1.2 Problems

All of the above treating time completely differently from space. Time was the single dependent variable w.r.t. which points in phase space evolved, while phase space was thought of as the collection of all physically distinct states a system could be in at any fixed time. In fact phase space need not have anything at all to do with space at all. We are tempted to think in terms of the mechanics of massive particles, where q and p essentially encode positions and velocities, but we could just as well study an Ising-type model where each q is the degree of excitation at a point rather than the location of a particle.

If we want to carry over the advantages of this formalism to the area of relativistic field theories, we ought to adapt it to treat space and time as equally as possible. To do so, we first observe that non-degeneracy of Ω and Ω^{-1} ensures that Hamiltonian time-evolution is well-defined in both directions, *i.e.*, reversible. So there exists a bijection between points in phase space and solutions to the equations of motion; after picking our phase space coordinates and deciding when to start our stopwatch, one such bijection is given by

$$(q, p) \mapsto (q(0), p(0)). \tag{18}$$

So from the start we can define our phase space to be the space of all on-shell solutions to our equations of motion. Since we want the whole picture to be manifestly covariant, those equations of motion will in turn arise from a Lagrangian action principle, and the Hamiltonian and symplectic form and so on will be derived objects.

2 Formalism

2.1 A Healthy Variational Principle

The ingredients needed to write down a field theory will include a d-dimensional spacetime M with boundary components $\partial M = \Gamma + \Sigma_+ - \Sigma_-$, fields ϕ^a, and some Lagrangian density $\mathcal{L}$ inducing an action functional

$$S_{\text{naive}}[\phi] = \int_M d^d x \mathcal{L}(\phi; x). \tag{19}$$

Notice that $\mathcal{L}$ is a spacetime d-form. For a local field theory, we typicallyl require that the value $\mathcal{L}(\phi; x)$ at some point depends on the field only through their values and derivatives at that same point, but we won't actually make use of that assumption here. We deduce the equations of motion by varying with respect to the fields, but if we do so using our naive action functional we find that (after integration by parts) the integrand has the undesirable form

$$\delta\mathcal{L} = E_a \delta\phi + d\Theta \implies \delta S_{\text{naive}} = \int_M E_a \delta\phi + \int_{\partial M} \Theta \tag{20}$$

For example, for a free field theory we find

$$\begin{aligned} \mathcal{L} &= \frac{1}{2} d\phi \wedge *d\phi = \frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi) \\ \implies \delta\mathcal{L} &= (\partial_\mu \delta\phi)(\partial^\mu \phi) \\ &= d(\delta\phi) \wedge *d\phi \\ &= -\delta\phi(d * d\phi) + d(\delta\phi \wedge *d\phi) \end{aligned} \tag{21}$$

where our equations of motion are

$$0 = d * d\phi \propto \partial^2 \phi. \tag{22}$$

This is undesirable in the sense that we want the equations of motion to be equivalent to the variational condition $\delta S = 0$. When we teach our undergraduate about action principles, we typically handwave and say that we require variations to die off fast enough to ignore total derivatives, but many subjects of modern interest (compact spaces, asymptotically flat spacetimes, large gauge transformations, etc) require that we be more careful.

To treat this more carefully, we start with boundary conditions. Here we will treat our spatial and timelike boundaries quite differently, and in defining our theory we will only prescribe boundary conditions along Γ. In this sense our prescription treats space and time in a nice joint way in the bulk, but distinguishes them on the boundary. The spatial boundary conditions will classically define what space of ϕ's we restrict to in our variational problem, and quantum-mechanically tell us what space to integrate over in our path integral. We think of future or past boundary conditions as defining states within some particular theory.

The simplest possible example is free field theory with Dirichlet boundary conditions. We know from experience that this is a healthy theory, while in our present framework we find it gives action variations

$$\begin{aligned} \delta S &= -\int_M \delta\phi(d * d\phi) + d(\delta\phi * d\phi) \\ &= \int_{\Gamma + \Sigma_+ - \Sigma_-} \delta\phi * d\phi \\ &= \int_{\Sigma_+ - \Sigma_-} \delta\phi * d\phi. \end{aligned} \tag{23}$$

Following this example, we find that even with the strictest and simplest boundary conditions it is too much to ask that a healthy variational principle take the literal form $\delta S = 0$; we must relax this and demand only

$$\text{boundary conditions} \implies \delta S = \int_M E_a \delta\phi^a + \int_{\Sigma_+ - \Sigma_-} \Psi \tag{24}$$

for some Ψ. Rather than writing things like "on-shell + boundary conditions $\implies$" everywhere, moving forward we will denote

$$\mathcal{C} \equiv \text{configuration space} \equiv \{\text{field histories } (\phi^a) \text{ obeying spatial BC's}\} \tag{25}$$

and

$$\tilde{\mathcal{P}} \equiv \text{pre-phase space} \equiv \{\text{field histories obeying spatial BC's + equations of motions}\} \subset \mathcal{C}. \tag{26}$$

Exercise: what is degree of Ψ in the de Rham complex on fields, and on spacetime? Answer: Ψ is a 1-form on fields and a $(D-1)$-form on spacetime.

In general, for boundary conditions less restrictive that the Dirichlet ones we just considered, we will need to supplement out bulk action with a compensating boundary term.

$$\begin{aligned} S &= \int_M \mathcal{L} + \int_{\partial M} \ell \\ \implies \delta S &= \int_M E_a \delta\phi^a + \int_{\partial M} (\Theta + \delta\ell). \end{aligned} \tag{27}$$

The most famous example of this strategy is the Gibbons-Hawking-York term that appears in classical GR. Again, we might naively think that we ought to reverse-engineer ℓ such that the boundary term here vanishes along the spatial boundary when the spatial BC's are obeyed.

$$\text{Naive: } \phi \in \mathcal{C} \overset{?}{\implies} (\Theta + \delta\ell)|_\Gamma = 0. \tag{28}$$

It turns out again that for our purposes (recall: our goal is that the on-shell variation should reduce to integrals over $\Sigma_\pm$) we can relax this to

$$\text{Better: } \phi \in \mathcal{C} \implies (\Theta + \delta\ell)|_\Gamma = dC|_\Gamma \tag{29}$$

for some spacetime $(D-2)-$form C. If this holds, then we have

$$\begin{aligned} \phi \in \mathcal{C} \implies \delta S &= \int_M E_a \delta\phi^a + \int_{\partial M} (\Theta + \delta\ell) \\ &= \int_M E_a \delta\phi^a + \int_{\Sigma_+ - \Sigma_-} (\Theta + \delta\ell) + \int_\Gamma dC. \end{aligned} \tag{30}$$

But we can rewrite by Stokes

$$0 = \int_{\partial M} dC = \int_{\Sigma_+ - \Sigma_-} dC + \int_\Gamma dC \tag{31}$$

so we have

$$\phi \in \mathcal{C} \implies \delta S = \int_M E_a \delta\phi^a + \int_{\Sigma_+ - \Sigma_-} (\underbrace{\Theta + \delta\ell - dC}_{\equiv \Psi}). \tag{32}$$

2.2 Defining a Symplectic Form

It turns out that this funny term Ψ appearing in the integrand will have enormous physical significance. First, for field histories in the pre-phase space we define the *pre-symplectic current* to be the field-space variation of Ψ.

$$\omega \equiv \text{pre-symplectic current} \equiv \delta\Psi|_{\tilde{\mathcal{P}}} = \delta(\Theta - dC)|_{\tilde{\mathcal{P}}}. \tag{33}$$

Then the pre-symplectic form we defined above will be exactly the integral of this object over any Cauchy slice Σ.

$$\tilde{\Omega} \equiv \text{pre-symplectic form} = \int_\Sigma \omega. \tag{34}$$

The reader should pause and verify that everything here type-checks: ψ is a spacetime $(D-1)$-form and a field 1-form, so ω is a spacetime $(d-1)$-form and a field 2-form, so $\tilde{\Omega}$ is a 2-form on $\tilde{\mathcal{P}}$.

By applying the boundary conditions and equations of motion we can verify that Ω has the properties we claimed a pre-symplectic form ought to have. First, it is closed as a form on $\tilde{\mathcal{P}}$ because the exterior derivative squares to 0.

$$\delta\Omega = \int_\Sigma \delta\omega = \int_\Sigma \delta^2\Psi = 0. \tag{35}$$

It is also independent of our choice of Cauchy surface. To see this, we observe that ω itself is closed as a spacetime form

$$d\omega = \delta d\Theta = \delta(\delta\mathcal{L} - E_a\delta\phi^a) = \delta E_a \wedge \delta\phi^a = 0 \tag{36}$$

where we have worked modulo the equations of motion.

3 Local Lagrangians

Assume we have fixed a d-dimensional spacetime M with boundary components $\partial M = \Gamma + \Sigma_+ - \Sigma_-$, fields ϕ^a, and compatible choices of L, ℓ, C and spatial boundary conditions, inducing equations of motion $E_a = 0$. Then varying the action gives

$$\delta S = \int_M E_a\delta\phi^a + \int_{\Sigma_+ - \Sigma_-} \Psi, \quad \Psi \equiv \Theta + \delta\ell - dC. \tag{37}$$

To interpret this object linear-algebraically, we introduce the configuration space

$$\mathcal{C} \equiv \text{configuration space} \equiv \{\text{field histories } (\phi^a) \text{ obeying spatial BC's}\} \tag{38}$$

which we think of as an infinite-dimensional manifold (though we won't attempt to rigorously interpret its smooth structure). Then S is a scalar field on $\mathcal{C}$ and δS its exterior derivative, so the latter can be thought of as a dual vector field. $\mathcal{C}$ has natural subspace

$$\tilde{\mathcal{P}} \equiv \text{pre-phase space} \equiv \{\text{field histories obeying spatial BC's + equations of motions}\}. \tag{39}$$

The prefix "pre-" indicates that this isn't quite the space of physical solutions, as we haven't yet quotiented out by gauge symmetries. We define the *pre-symplectic current* by taking the field-space exterior derivative of Ψ and restricting the result to take in vectors tangent to $\tilde{\mathcal{P}}$.

$$\omega \equiv \text{pre-symplectic current} \equiv \delta\Psi|_{\tilde{\mathcal{P}}} \tag{40}$$

ω is a 2-form field over $\tilde{\mathcal{P}}$ and a spacetime $(d-1)$-form field, which is local in the sense that if we expand in the natural basis

$$\omega = \int_M d^dx \int_M d^dy\, \omega_{ab}(\phi, x, y)\delta\phi^a(x) \wedge \delta\phi^b(y) \tag{41}$$

the components $\omega_{ab}(\phi, x, y)$ depends only on the germs of ϕ at x, y. Then given any Cauchy slice Σ (roughly, a codimension-1 submanifold of M diffeomorphic to $\Sigma_+ \simeq \Sigma_-$ whose causal past and future cover the complement $M - \Sigma$) we define the pre-phase space 2-form

$$\tilde{\Omega} \equiv \text{pre-symplectic form} \equiv \int_\Sigma \omega. \tag{42}$$

One can check explicitly that this is independent of the choice of Σ.

4 Covariant Lagrangians

Given a vector field of M which we think of as infinitesimal, the corresponding induced variation on an arbitrary dynamical tensor field ϕ is given as usual by the Lie derivative

$$\delta_\xi\phi = \mathcal{L}_\xi\phi. \tag{43}$$

If this variation respects the spatial boundary conditions, we can write the action on the dynamical fields ϕ^a of our theory in terms of a vector field along $\mathcal{C}$

$$X_\xi = \int_M d^d x \left(\mathcal{L}_\xi \phi^a(x)\right) \frac{\delta}{\phi^a(x)}. \tag{44}$$

Then individual fields vary as

$$\delta_\xi \phi^a(x) = X_\xi \phi^a(x). \tag{45}$$

More generally, given some scalar field f built from our dynamical fields, we have

$$\delta_\xi f = X_\xi f. \tag{46}$$

Crucially, we interpret δ_ξ as the change in f induced by slightly changing every dynamical field ϕ^a, but *not* changing any background fields ξ^b. A theory is said to be *covariant* w.r.t. the spacetime vector field ξ (or more often, w.r.t. some algebra of such vector fields, or a group of non-infinitesimal spacetimes diffeos which linearize to the former) if these variations respect the boundary conditions and satisfaction of the equations of motion. This is non-trivial exactly because we are varying the dynamical fields but not the background fields; if we were varying both, then covariant under the full diffeomorphism group would be automatic. Conversely, under our definitions the background fields can be thought of as breaking full diffeomorphism invariance down to some subgroup.

5 Diffeomorphism Charges

Consider some infinitesimal diffeomorphism ξ and corresponding configuration-space vector field X_ξ, w.r.t. which our theory may or may not be covariant. We want to find a corresponding scalar function H_ξ on pre-phase space such that

$$\delta H_\xi = -\tilde{\Omega}(X_\xi, \cdot). \tag{47}$$

$\tilde{\Omega}$ generally has zero-modes, *i.e.*, configuration space vector fields $\tilde{X}$ obeying

$$\tilde{X} \cdot \tilde{\Omega} \equiv \tilde{\Omega}(\tilde{X}, \cdot) = 0. \tag{48}$$

Such zero-modes generate flows in pre-phase space $\tilde{\mathcal{P}}$ which we identify with gauge orbits, so that the true phase space $\mathcal{P}$, or equivalently the set of physically distinct solutions, is the quotient of $\tilde{\mathcal{P}}$ by the group $\tilde{G}$ of gauge transformations $\tilde{\mathcal{P}} \to \tilde{\mathcal{P}}$ generated by the zero-modes.

$$\mathcal{P} \equiv \text{phase space} = \{\text{physically distinct solutions} = \text{gauge orbits}\} = \tilde{\mathcal{P}}/\tilde{G}. \tag{49}$$

We will assume this quotient is smooth enough that $\mathcal{P}$ is analytically well-behaved, with tangent vectors we identify as vectors tangent to $\tilde{\mathcal{P}}$ modulo addition of zero-modes. In particular, even thought δH_ξ is a 1-form field on $\tilde{\mathcal{P}}$, the fact that it annihilates 0-modes means it is equally well-defined as a 1-form field on $\mathcal{P}$; following [?] we will abuse notation and write δH_ξ for both 1-forms. Similarly, $\tilde{\Omega}$ is well-defined as an equivalent 2-form over $\mathcal{P}$, though when we interpret it as such we write it as Ω rather than $\tilde{\Omega}$.

Like any quadratic form, the phase-space 2-form $\tilde{\Omega}$ can be thought of as a map from vectors to 1-forms

$$Y \in T\mathcal{P} \overset{\Omega}{\mapsto} (X \mapsto \Omega(X, Y)) \in T^*\mathcal{P} \tag{50}$$

or by mild abuse of notation

$$\Omega(Y)(X) \equiv \Omega(X, Y) \text{ or } \Omega(Y) \equiv \Omega(\cdot, Y) \tag{51}$$

Note that the order in which we fill in the slots of Ω is purely convention, but does give us minus signs we must track. This map is invertible, with inverse equivalently describable as a phase space bivector Ω^{-1} acting as

$$\alpha\left(\Omega^{-1}(\beta)\right) \equiv \Omega^{-1}(\alpha, \beta) \text{ or } \Omega^{-1}(\beta) \equiv \Omega^{-1}(\cdot, \beta). \tag{52}$$

Observe that Ω, Ω^{-1} are both defined so that the right-most argument is "filled in first." Ω^{-1} being the left-inverse of Ω^{-1} has equivalent phrase in the language of quadratic forms as

$$\begin{aligned} X &= \Omega^{-1}(\Omega(X)) \\ &= \Omega^{-1}(\Omega(\cdot, X)) \\ &= \Omega^{-1}\left(\cdot, \Omega(\cdot, X)\right). \end{aligned} \tag{53}$$

This is slightly clearer if we let a 1-form eat X, so that variously

$$\begin{aligned} \alpha(X) &= \alpha(\Omega^{-1}(\Omega(X)) \\ &= \alpha(\Omega^{-1}(\Omega(\cdot, X))) \\ &= \alpha(\Omega^{-1}(\cdot, \Omega(\cdot, X))) \\ &= \Omega^{-1}(\alpha, \Omega(\cdot, X)) \\ &= \Omega^{-1}(\alpha, \Omega(X)). \end{aligned} \tag{54}$$

In particular, let the relevant vector field be $X = X_\xi$ and let the 1-form be $\alpha = \delta f$, the exterior derivative of some scalar field on phase space. Then the variation of this scalar field under the phase space diffeomorphism X_ξ is exactly

$$\begin{aligned} X_\xi(f) &= \delta f(X_\xi) \\ &= \Omega^{-1}(\delta f, \Omega(X_\xi)) \\ &= \Omega^{-1}(\delta f, \Omega(\cdot, X_\xi)) \\ &= \Omega^{-1}(\delta f, \delta H_\xi). \end{aligned} \tag{55}$$

It is in this sense that δH_ξ interacts with the symplectic form to generate the changes in observables (concretely, scalars on phase space) induced by phase space diffeomorphisms. Now we want to construct H_ξ, starting by showing that the phase space 1-form $\Omega(X_\xi)$ is indeed exactly an exterior derivative. To do so, we introduce the Noether current associated to ξ

$$J_\xi \equiv X_\xi \cdot \Theta - \iota_\xi L. \tag{56}$$

Here we write $\cdot$ for contraction along field arguments and ι for contraction along spacetime arguments. The result is a spacetime $(d-1)$-form and a scalar on pre-phase space. Now assume that L is covariant under ξ, in the sense that it is built from the dynamical field ϕ^a and background fields χ^b in such a way that its dynamical field variation under ξ (given by the action of the configuration space vector field $X_\xi L$) agrees with its naive variation under the spacetime diffeomorphism (given by the spacetime Lie derivative $\delta_\xi L \equiv \mathcal{L}_\xi L$). Then we have

$$dJ_\xi = d\left(X_\xi \cdot \Theta\right) - d\left(\iota_\xi L\right). \tag{57}$$

Since Θ is a spacetime $(d-1)$-form while X_ξ is a spacetime constant, the first spacetime exterior derivative acts as

$$\begin{aligned} d(X_\xi \cdot \Theta) &= X_\xi \cdot d\Theta \\ &= X_\xi \cdot (\delta L - E_a \delta\phi^a) \\ &= X_\xi(L). \end{aligned} \tag{58}$$

Meanwhile, applying Cartan's magic formula to spacetime contractions and spacetime exterior derivatives and recalling that L is a spacetime top form, we have

$$d(\iota_\xi L) = \mathcal{L}_\xi L - \iota_\xi(dL) = \mathcal{L}_\xi L. \tag{59}$$

Covariance of L w.r.t. ξ is exactly the assumption that these two terms agree, so the difference vanishes.

$$dJ_\xi = X_\xi(L) - \mathcal{L}_\xi L = 0. \tag{60}$$

With this in hand, we decompose

$$\delta H_\xi = -X_\xi \cdot \Omega = -X_\xi \cdot \int_\Sigma \omega, \quad \omega \equiv \delta(\Omega + \delta\ell - dC) = \delta\Theta - d\delta C \tag{61}$$

and compute the integrand over Σ as

$$\begin{aligned}
-X_\xi \cdot \omega &= -X_\xi \cdot \delta(\Theta - dC) \\
&= \delta(X_\xi \cdot (\Theta - dC)) - \mathcal{L}_{X_\xi}(\Theta - dC) \text{ (Cartan magic)} \\
&= \delta(X_\xi \cdot \Theta) - \delta(X_\xi \cdot dC) - \mathcal{L}_{X_\xi}\Theta + \mathcal{L}_{X_\xi} dC \text{ (expand)} \\
&= \delta J_\xi + \iota_\xi \delta L - \delta(X_\xi \cdot dC) - \mathcal{L}_{X_\xi}\Theta + \delta_\xi dC \text{ (definitions of } J_\xi,\ \delta_\xi) \ .
\end{aligned} \tag{62}$$

Now if L is covariant, so is $\delta L = d\Theta$ on-shell, hence so is (plausibly) Θ itself[1].

$$\begin{aligned}
\implies -X_\xi \cdot \omega &= \delta J_\xi + \iota_\xi \delta L - \delta(X_\xi \cdot dC) - \mathcal{L}_\xi \Theta + \delta_\xi dC \\
&= \delta J_\xi + \iota_\xi \delta L - \mathcal{L}_\xi \Theta + d\left(\delta_\xi C - \delta(X_\xi \cdot C)\right) \\
&= \delta J_\xi + \iota_\xi (E_a \delta\phi^a + d\Theta) - \mathcal{L}_\xi \Theta + d\left(\delta_\xi C - \delta(X_\xi \cdot C)\right) \\
&= \delta J_\xi + \iota_\xi d\Theta - \mathcal{L}_\xi \Theta + d\left(\delta_\xi C - \delta(X_\xi \cdot C)\right) \\
&= \delta J_\xi + d\left(\delta_\xi C - \delta(X_\xi \cdot C) - i_\xi \Theta\right) .
\end{aligned} \tag{63}$$

So $-X_\xi \cdot \omega$ is the sum of a field-exact object and a spacetime-exact one. Integrating over a Cauchy slice then gives

$$\begin{aligned}
\delta H_\xi &= -X_\xi \cdot \tilde{\Omega} \\
&= -\int_\Sigma X_\xi \cdot \omega \\
&= \int_\Sigma \delta J_\xi + \int_{\partial\Sigma} \left(\mathcal{L}_\xi C - \delta(X_\xi \cdot C) - i_\xi \Theta\right) \\
&= \int_\Sigma \delta J_\xi + \int_{\partial\Sigma} \left(\iota_\xi dC + d(\iota_\xi C) - \delta(X_\xi \cdot C) - i_\xi \Theta\right) \\
&= \delta\left(\int_\Sigma J_\xi + \int_{\partial\Sigma} (\iota_\xi \ell - X_\xi \cdot C)\right) .
\end{aligned} \tag{64}$$

So we are done, and we can identify

$$H_\xi \equiv \int_\Sigma J_\xi + \int_{\partial\Sigma} (\iota_\xi \ell - X_\xi \cdot C) + (\text{const}) \tag{65}$$

where the additive shift is a constant c-number which we typically set to 0.

5.1 Independence of Cauchy Surface

Now we check that $H_\xi(\Sigma)$ is in fact independent of the choice of Σ. So consider some other choice Σ', so that the difference is purely spatial and takes the form

$$\partial\Sigma - \partial\Sigma' = \partial\Xi \subset \Gamma \text{ with } \Xi \subset \Gamma. \tag{66}$$

Then the difference between H_ξ defined along the two Cauchy surfaces is

$$\begin{aligned}
H_\xi(\Sigma) - H_\xi(\Sigma') &= \int_{\Sigma - \Sigma'} \left(J_\xi + d\left(\iota_\xi \ell - X_\xi \cdot C\right)\right) \\
&= \int_{\Sigma - \Sigma'} J_\xi + \int_{\partial\Xi} (\iota_\xi \ell - X_\xi \cdot C) \\
&= \int_{\Sigma - \Sigma'} J_\xi + \int_{\Xi} d\left(\iota_\xi \ell - X_\xi \cdot C\right)
\end{aligned} \tag{67}$$

To simply the integral against J_ξ, we denote by $\mathcal{M} \subset M$ the open region enclosed by Σ, Σ', Ξ. Carefully tracking orientations and signs, we then have

$$\partial\mathcal{M} = \Sigma - \Sigma' - \Xi \tag{68}$$

[1]In our original construction Θ was only defined modulo shifts by a spacetime-exact top form, so the real claim is that Θ can be chosen to be covariant.

where we observe that the relative signs can be deduced by enforcing $\partial^2\mathcal{M}=0$. Then since $dJ_\xi=0$ on pre-phase space we can rewrite

$$\int_{\Sigma-\Sigma'} J_\xi = \underbrace{\int_{\partial\mathcal{M}} J_\xi}_{\to 0} + \int_\Xi J_\xi \tag{69}$$

and we have neatly

$$\begin{aligned} H_\xi(\Sigma) - H_\xi(\Sigma') &= \int_\Xi \left(J_\xi + d\left(\iota_\xi \ell - X_\xi\cdot C\right)\right) \\ &= \int_\Xi \left(X_\xi\cdot\Theta - \iota_\xi\cdot L + d\left(\iota_\xi\ell - X_\xi\cdot C\right)\right) \\ &= \int_\Xi \left(X_\xi\cdot(\Theta - dC) - \iota_\xi\cdot L + d(\iota_\xi\ell)\right) \\ &= \int_\Xi \left(-X_\xi\cdot\delta\ell - \iota_\xi\cdot L + d(\iota_\xi\ell)\right) \end{aligned} \tag{70}$$

(TODO)

6 Examples

6.1 Scalar Field

Consider the theory of a single scalar

$$L = -\left(\frac{1}{2}(\nabla_\mu\phi)(\nabla^\mu\phi) + V(\phi)\right)\epsilon. \tag{71}$$

Variation of the Lagrangian gives

$$\begin{aligned} \delta L &= -\left((\nabla^\mu\phi)\delta\nabla_\mu\phi + V'(\phi)\delta\phi\right)\epsilon \\ &= \left(\nabla^\mu\nabla_\mu\phi - V'(\phi)\right)\epsilon\delta\phi - \nabla_\mu(\delta\phi\nabla^\mu\phi)\epsilon \\ &\equiv E\delta\phi - d\Omega \end{aligned} \tag{72}$$

where by comparison we identify

$$E = (\nabla^2\phi - V'(\phi))\epsilon \tag{73}$$

and

$$d\Theta = -\nabla_\mu(\delta\phi\nabla^\mu\phi)\epsilon = -d\left((\delta\phi\nabla\phi)\cdot\epsilon\right) \impliedby \Theta = -\delta\phi(\nabla\phi)\cdot\epsilon \equiv \theta\cdot\epsilon. \tag{74}$$

At this point we are giving up on $\cdot$ vs ι and write both kinds of contractions with $\cdot$, leaving disambiguation to context clues. ℓ can only be determined jointly with our boundary conditions, which require that

$$(\delta\ell + \Theta)|_\Gamma = dC \tag{75}$$

be an exact form. The spacetime $(d-1)$-form Θ restricts to Γ as

$$\Theta|_\Gamma = n_\mu\theta^\mu\epsilon_{\partial M} = n_\mu\delta\phi\nabla^\mu\phi\epsilon_{\partial M} \tag{76}$$

where n is the unit normal form to ∂M and $\epsilon_{\partial M}$ is the induced volume form on ∂M. We can consistently take the simplest possible case with ℓ, C vanishing as long as $\Theta|_\Gamma$ vanishes, for which it suffices to take either Dirichlet

$$\delta\phi|_\Gamma = 0 \tag{77}$$

or Neumann

$$n^\mu\nabla_\mu\phi|_\Gamma = 0 \tag{78}$$

boundary conditions. Now we have our pre-symplectic current

$$\begin{aligned}\omega &= \delta\left(\Theta + \delta\ell - dC\right)|_{\tilde{\mathcal{P}}} \\ &= \delta\Theta|_{\tilde{\mathcal{P}}} \\ &= -\delta\left(\delta\phi(\nabla\phi)\cdot\epsilon\right) \\ &= -(\nabla\delta\phi)\cdot\epsilon\wedge\delta\phi \\ &\equiv \hat{\omega}\cdot\epsilon\end{aligned} \tag{79}$$

where

$$\hat{\omega}^\mu = -(\nabla^\mu\delta\phi)\wedge\delta\phi \tag{80}$$

and where the wedge product $\wedge$ is taken in the complex over pre-phase space. Integrating this over a Cauchy surface Σ gives the pre-symplectic form

$$\begin{aligned}\tilde{\Omega} &= -\int_\Sigma \hat{n}_\mu\hat{\omega}^\mu\epsilon_\Sigma \\ &= -\int_\Sigma \hat{n}^\mu(\nabla_\mu\delta\phi)\wedge\delta\phi\epsilon_\Sigma\end{aligned} \tag{81}$$

where Σ has induced volume form ϵ_Σ and unit normal $\hat{n}$. We see that this is non-degenerate, so there are no zero-modes by which we must quotient and we have

$$\mathcal{P} = \tilde{\mathcal{P}}, \quad \Omega = \tilde{\Omega}. \tag{82}$$

Now given ξ we can cook up a current

$$J_\xi = X_\xi\cdot\Theta - \xi\cdot L \tag{83}$$

where

$$X_\xi = \int_M d^dx(\xi\cdot d\phi)(x)\frac{\delta}{\delta\phi(x)} \tag{84}$$

so the first term is

$$\begin{aligned}X_\xi\cdot\Theta(x) &= \left(\int_M d^dx(\xi\cdot d\phi)(y)\frac{\delta}{\delta\phi(y)}\right)\cdot\left(-(\nabla\phi(x))\cdot\epsilon\delta\phi(x)\right) \\ &= -(\mathcal{L}_\xi\phi)(x)(\nabla\phi\cdot\epsilon)(x) \\ &= -\xi_\nu(\nabla^\mu\phi)(\nabla^\nu\phi)\partial_\mu\cdot\epsilon\end{aligned} \tag{85}$$

while the second term gives

$$\begin{aligned}-\xi\cdot L &= \frac{1}{2}\xi\cdot\left[\left((\nabla_\mu\phi)(\nabla^\mu\phi) + V(\phi)\right)\epsilon\right] \\ &= \xi_\nu g^{\mu\nu}\left[\frac{1}{2}(\nabla_\alpha\phi)(\nabla^\alpha\phi) + V(\phi)\right]\partial_\mu\cdot\epsilon.\end{aligned} \tag{86}$$

We can therefore write more nicely

$$J_\xi = j_\xi\cdot\epsilon \tag{87}$$

where

$$j_\xi^\mu = -\xi_\nu\left((\nabla^\mu\phi)(\nabla^\nu\phi) - g^{\mu\nu}\left[\frac{1}{2}(\nabla_\alpha\phi)(\nabla^\alpha\phi) + V(\phi)\right]\right). \tag{88}$$

6.2 Maxwell

Free Maxwell theory has Lagrangian

$$L = -\frac{1}{2}F\wedge *F, \quad F = dA \tag{89}$$

which in turn has variation

$$\begin{aligned}\delta L &= -\delta F\wedge *F \\ &= -\delta A\wedge d*F - d(\delta A\wedge *F) \\ &\equiv \delta A_\mu E^\mu + d\Theta\end{aligned} \tag{90}$$

from which we read off

$$\Theta = -\delta A \wedge *F. \tag{91}$$

If we apply Dirichlet boundary conditions

$$\delta A|_\Gamma = 0 \tag{92}$$

then we can consistently set ℓ and C to 0. Then our pre-symplectic current and forms become

$$\omega = \delta\Theta = -\delta A \wedge *\delta F \implies \tilde{\Omega} = -\int_\Sigma \delta A \wedge *\delta F. \tag{93}$$

This has nontrivial zero modes, but the latter depend on our chosen boundary conditions. A candidate zero mode has general form

$$\tilde{X} = \int_M d^d x V_\mu(x) \frac{\delta}{\delta A_\mu}. \tag{94}$$

To compute its contraction against $\tilde{\Omega}$ we use coordinates (t, y) for M and x for Σ, so that

$$\begin{aligned}
\tilde{X} \cdot \tilde{\Omega} &= \left(\int_M d^d y V_\mu(y) \frac{\delta}{\delta A_\mu(y)}\right) \cdot \left(\int_\Sigma \delta A(x) \wedge *d\delta A(x)\right) \\
&= \int_\Sigma \int_M d^d y V_\mu(y) \frac{\delta}{\delta A_\mu(y)} \cdot (\delta A \wedge *d\delta A) \\
&= \int_\Sigma (V \wedge *d\delta A - \delta A \wedge *dV).
\end{aligned} \tag{95}$$

If V is an exact form $V = d\lambda \implies dV = 0$, then the second term vanishes and integration by parts lets us rewrite the first as

$$\begin{aligned}
\tilde{X} \cdot \tilde{\Omega} &= \int_\Sigma d\lambda \wedge *d\delta A \\
&= \int_{\partial\Sigma} \lambda * \delta F.
\end{aligned} \tag{96}$$

Notice that this does not vanish in general! Our Dirichlet boundary conditions force

$$A|_\gamma \text{ fixed } \implies \lambda|_\Gamma = \text{ const} \tag{97}$$

but the vanishing of $\tilde{X} \cdot \tilde{\Omega}$ identically on pre-phase space happens iff $\lambda|_\Gamma = 0$. So we have rediscovered the classical notion of a large gauge transformation: a transformation $A \to A + d\lambda$ on a spacetime $M \simeq \mathbb{R} \times \Sigma$ for Σ compact should only be thought of as a bona fide redundancy in our enumeration of physical states when λ vanishes along the spatial boundary; otherwise the transformation takes us to a physically distinguishable new state.

7 TODO

Follow-up work:

- BH entropy: [**?**, **?**, **?**]
- Nice notes on corner proposal: [**?**]